A

THE UNITED REPUBLIC OF TANZANIA.
PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT.

LAKE ZONE FORM FOUR MOCK EXAMINATION

033/1

BIOLOGY 1.

MARKING SCHEME.

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1. SECTION A 16 MARKS

Question NO	(i)	ii	iii	iv	v	vi	vii	viii	ix	x
Answer	C	c	C	A	D	A	B	D	C	A

01 mark @ = 10 Marks

2.

Question NO	(i)	(ii)	(iii)	(iv)	(v)	(vi)
Answer	D	C	H	A	B	E

01 mark @ = 06 Marks

SECTION B 54 MARKS.

3. (a) Starch ——(01 mark)

(b) Natural sources

- Cassava.
- Sweet potatoes
- Wheat
- Millet
- Sorghum
- Maize
-

Any 04 pts @ 01 = 04 Marks

3 (c) Functions of starch

(i) Major sources of energy

(ii) Major component of cell membrane

01 Mark @ = 02 Marks

3(d) Parts where the food is digested

(i) Mouth - - - 01 mark

(ii) Deudenum - - - 01 mark

TOTAL = 09 MARKS

4 (a) Specialized plant cells its functions

(i) Guard cells

- Controls opening and closing of the stomata

(ii) Root hair cells

- Absorption of water and Mineral salts from the soil

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(iii) Pelisade cells

- It contains chloroplasts for photosynthesis

(iv) Xylem

- ~~Absorption~~ Conduct water from the roots to the roots.

(v) Phloem

- Conducts food from the leaves to storage organs.

Any 3 pts @ 2 Marks @ = 06 Marks

(b) Differences between animal and plant cells.

Plant cell	Animal cell.
Has cell wall Has chloroplasts It is regular in shape It has large and permanent vacuole plants stores food in the form of starch	~~Anim~~ Has no cell wall Lack chloroplasts. It is irregular in shape. - It has small and tempurary vacuole - Store food in the form of glycogen

Any 03 pts @ 01 = 03 Marks

05. Forms of Asexual Reproduction

(i) Fission.

Is the form of asexual reproduction in which a parent cell divides into two or more individuals of equal size Example, binary and multiple fission.

(ii) Fragmentation.

Is the form of asexual reproduction where by an organism breaks down into fragments or pieces and each fragment develop into mature organisms example: Tapeworms.

(iii) Spore formation (sporulation)

Is the form of asexual reproduction where by an organism reproduces new organisms by using spores as the reproductive structure.

(iv) Budding

Is the form of asexual reproduction in which a new individual is produced as an outgrowth of the parent.

Example: Yeast.

(v) Vegetative propagation

Is the form of asexual reproduction which occurs when a new plant grows from the fragment of the parent plants or through specialized of reproductive structures such as roots, leaves and stem.

Any 3 pts @ 03 Marks

TOTAL = 09 Marks.

06. (a) Importance of gaseous exchange in plants

(i) It enable plants to obtain CO_2 gas which is the raw material for photosynthesis

(ii) It enable plants to obtain oxygen gas which is essential for production of energy.

(iii) Enables the plant to remove the waste products such as $CO_{2(g)}$ during the night.

Marks $01\frac{1}{2}$ @ $= 04\frac{1}{2}$ Marks

(b) Adaptation of Respiratory Surfaces.

(i) They are thin in order to reduce the diffusion distance

(ii) They are moist in order to dissolve gases that diffuses in a solution form.

(iii) They are highly branched, folded, or flattened in order to increase surface area for gaseous exchange

(iv) They are surrounded by blood capillaries so that gases can be taken to and from the cells easily.

(v) They are well ventilated so that the gases can pass through them easily.

3 pts @ $0\frac{1}{2}$ = $04\frac{1}{2}$ Marks

TOTAL = 09 Marks

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7. Secondary sexual characteristics in Boys during puberty.

(i) Enlargement of testes and penis
(ii) Production of sperms begins
(iii) Development of a deep and Course voice
(iv) Development of hairs under armpits and pubic regions.
(v) Wet dreams begins

Any 3pts @ $01\frac{1}{2}$ = $04\frac{1}{2}$ Marks

Secondary sexual characteristics in Girls During puberty.

(i) Development and increase of breast size
(ii) Beginning of ovulation and menstruation
(iii) Broadening of hips and pelvic girdles
(iv) Development of soft and higher pitched voice
(v) Increase of secretion of mucus from the vagina.

Any 3pts @ $01\frac{1}{2}$ = $04\frac{1}{2}$ Marks

TOTAL = 09 Marks.

8. Let I^A = genes represent blood group A.
I^B = genes represent blood group B (01)
I^O = genes represent blood group O

But, a mother to have the child of blood group O must be heterozygous.

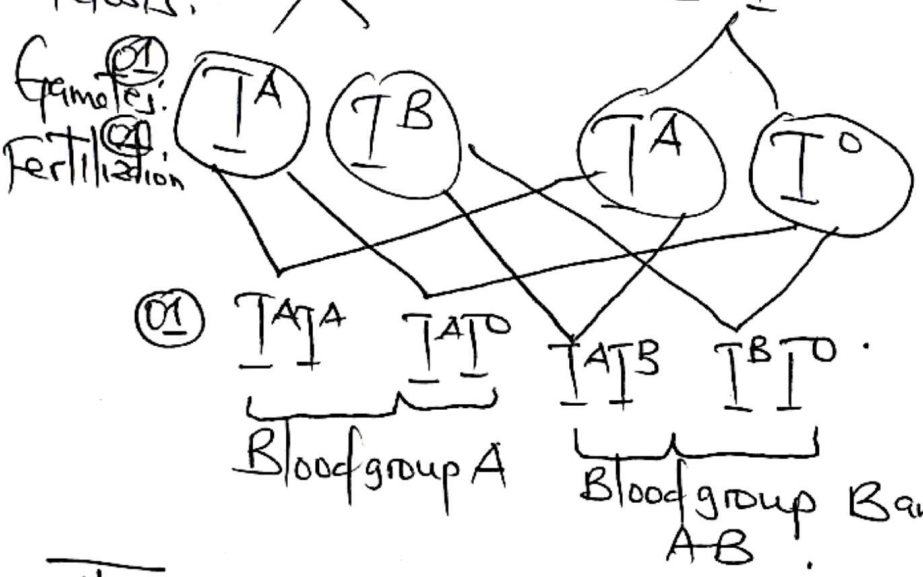
Parental phenotype: Blood group AB father × Blood group A mother (01)

Parental genotype: I^AI^B × I^AI^O. (01)

Meiosis: (01)

Gametes: (01)

Fertilization (01)



(01) I^AI^A I^AI^O I^AI^B I^BI^O.

Blood group A — Blood group B and AB.

Comment: The woman's claims was (02) incorrect, therefore the father of blood group AB was not the Biological father of her son.

TOTAL = 09 Marks.

09. Introduction

Meaning of poikilotherms
Mentioning some examples of poikilotherms — 01 1/2 Marks

Main Body

Behavioural means of regulation of body Temperature in poikilotherms.

(i)– Sun basking
(ii)– Hibernation
(iii)– Aestivation
(iv)– Migration
(v)– Huddling

Point --- 01 mark
Explanation---02Marks
Any 4pts@03=12Marks

Conclusion:

Any relevant Conclusion - - - - 01/2 Marks

TOTAL = 15 Marks

10.

Introduction 01½ marks

Explain about Mammals with examples...

Main Body

Economic importance of Mammals.

Advantages

(i) They are used as the source of food to human and other animals. Eg Cow produces milk and meat.

(ii) They produce organic manure through exretion of feaces and dungs

(iii) Products from Mammals have Commercial values as used to make different products Example, Animal skin used for making leather

(iv) Some Mammals are used in scientific studies and research. Example Rats that are trained to detect explosives

(v) Some animals are used for transportation and Cultivation of crops. Example Cows and Donkeys.

(vi) Some Mammals are used in security activities Example. Dogs

(vii) Production of Bio-gas

Any 3pts @ 02 = 06 Marks

10.

Disadvantages of Mammals

(i) Some Mammals are dangerous to human and other organisms
Example, Lion, Leopards, buffallo etc.

(ii) Some are vectors of diseases
Example Rats and cats that spread fleas which leads to plague.

(iii) Some mammals are pests to crops
Example, Monkeys, Rats and pigs.

(iv) Some ruminant mammals produces methane gas which contributes to global warming.

Any 3pts @ 02 = 06marks

Conclusion

Any relevant conclusion - - - 01½ Marks

TOTAL = 15Marks.

11. Introduction

Meaning of Body defense system 0½ Marks

Main Body

Body defensive system

(i) Eyes
- Production teaRs
- function of eyelids
- Function of eyelashes and eye brows

(ii) SKIN
- Act as the barrier
- sweat glands to produce sweat
- Saboceous glands to produce sebum

(iii) EARS
- Productions of waxy materials

Any 4pts @3marks = 12Marks

(iv) DIGESTIVE SYSTEM
- Production of saliva from the mouth
- Production of juices from the stomach, ileum eg ~~H~~ Hydrochloric acids.

(v) BLOOD
- Contains antibodies eg WBCs

Conclusion

Any relevant conclusion 0½ Marks

TOTAL = 15 Marks